

Partial correlation between literacy rate and number of crimes committed given poverty rate

```
# Loading the data set:-
```

```
x = read.delim("C:/Users/rahul/OneDrive/Documents/ALL.txt")
```

```
# Knowing some primary things about the data set:-
```

```
dim(x)
```

```
## [1] 34 4
```

```
names(x)
```

```
## [1] "State.UT"
```

```
## [2] "Total.Crime.against.Women..IPC..SLL."
```

```
## [3] "Literacy.rate"
```

```
## [4] "Poverty.rate"
```

```
# Computing Partial correlation:-
```

```
f1 = lm(Total.Crime.against.Women..IPC..SLL.~Poverty.rate , data = x) # Regress Total.Crime.against.Wom
```

```
f2 = lm(Literacy.rate~Poverty.rate , data = x) # Regress Literacy.rate on Poverty.rate
```

```
f11 = f1$residuals # Taking residuals(true value - estimated value) of f1
```

```
f12 = f2$residuals # Taking residuals(true value - estimated value) of f2
```

```
r = cor(f11,f12) # Partial correlation between literacy rate and number of crimes committed
```

```
r # Print required partial correlation
```

```
## [1] -0.05489115
```